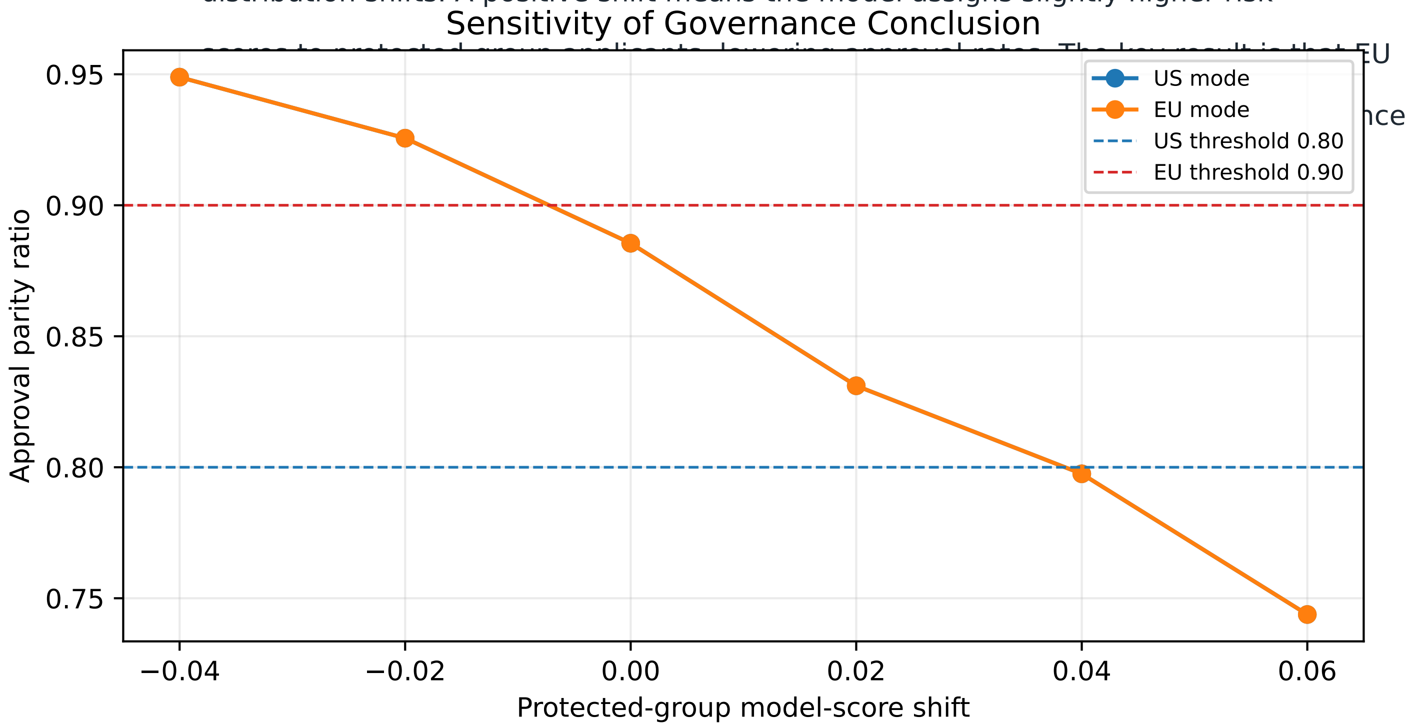
Task 3 - Sensitivity Analysis

This page tests how much the conclusion changes if the protected-group score distribution shifts. A positive shift means the model assigns slightly higher risk scores to protected-group applicants, lowering approval rates. The key result is that EU

nce



Selected scenarios

protected_group_score_shift	active_jurisdiction	approval_parity_ratio	threshold	fairness_flag
0.00	US	0.8855	0.8	False
0.00	EU	0.8855	0.9	True
0.02	US	0.8311	0.8	False
0.02	EU	0.8311	0.9	True
0.04	US	0.7975	0.8	True
0.04	EU	0.7975	0.9	True

Interpretation: the conclusion is not a fragile one-line claim. The tool makes the threshold choice visible, so management can see when a small distribution shift changes the required governance action.

Task 3 - Sensitivity Interpretation

Why this matters

A fairness conclusion should not depend on a hidden modeling assumption. The sensitivity test makes the threshold boundary visible by changing the protected-group score distribution in small increments and recalculating the approval parity ratio. This is not meant to predict a real macroeconomic shock; it is a governance stress test that asks whether the conclusion is robust near the rule threshold.

What changes across jurisdictions

At the observed parity ratio of 0.885, US mode is compared with the internal 0.80 review threshold, while EU mode is compared with Meridian's internal 0.90 high-risk AI governance threshold. The EU threshold is not statutory; it is a conservative escalation setting attached to separate Article 10, 14 and 17 controls.

Management implication

The result tells senior management that the product should not present a single global fairness conclusion. The same model output requires different governance evidence, escalation language and human oversight depending on the active jurisdiction. A future production version should automate this sensitivity test monthly and attach the result to model-risk committee materials.

Limitation of the sensitivity analysis

The analysis changes model scores synthetically rather than retraining the model on new real-world data. It is useful for governance reasoning, but not a substitute for production drift monitoring, challenger models, legal review or customer remediation analysis.